Master Of Data Analytics

Winter 2026 Capstone Project (DAMO-699-2)

Capstone Project Proposal:

Real-Time Traffic Congestion Prediction and Operational Optimization in Toronto Using Advanced Data Analytics

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Project Idea:

Rapid urbanization and population growth have led to increasing traffic congestion in major metropolitan cities, resulting in economic inefficiencies, increased greenhouse gas emissions, and reduced quality of life. Toronto, as one of Canada's largest and busiest cities, experiences persistent congestion that affects commuters, public transportation systems, and urban mobility planning. Conventional traffic management approaches often rely on static rules or historical averages, limiting their ability to respond proactively to dynamic and rapidly changing traffic conditions.

This capstone project proposes the development of a data-driven traffic analytics system that integrates historical traffic data with real-time traffic and weather information to predict short-term congestion and support operational decision-making. By applying advanced data analytics, forecasting techniques, and operations analytics, the project aims to demonstrate how predictive insights and optimization scenarios can improve traffic flow management. Interactive dashboards and a supporting web-based interface will be used to visualize insights and communicate results effectively to transportation planners and decision-makers. Prior research has demonstrated the effectiveness of spatio-temporal models for short-term traffic and urban mobility prediction (Zhang et al., 2017).

Problem Identification:

Traffic congestion in Toronto contributes to longer travel times, higher fuel consumption, increased emissions, and significant economic costs. Despite the availability of real-time traffic data through modern sensing technologies and APIs, traffic management systems often lack predictive capabilities that allow planners to anticipate congestion before it occurs. As a result, many interventions are reactive rather than proactive.

The central problem addressed in this project is the absence of an integrated, analytics-driven decision support system that combines historical traffic patterns with real-time data to predict congestion and evaluate operational interventions. Addressing this gap can enable data-driven planning, improved signal timing strategies, and more efficient use of transportation infrastructure.

In this study, traffic congestion is treated as a continuous quantitative variable, rather than a binary condition. Congestion will be measured using traffic-related indicators such as traffic volume, estimated delay, and speed reduction at intersections or road segments. Accordingly, the problem is formulated as a short-term traffic forecasting problem, rather than a classification task. This formulation enables the application of time-series and regression-based analytical models to predict congestion levels and evaluate operational strategies under different scenarios.

Aim of Investigation:

The primary aim of this project is to investigate whether historical traffic data combined with real-time traffic and weather information can be used to accurately predict short-term traffic congestion in Toronto and support operational optimization strategies.

Research Questions and Hypotheses:

Research Question 1:

Do models trained on historical traffic data outperform naïve baseline models in short-term traffic congestion forecasting?

Null Hypothesis (H_{01}): Models using historical traffic data do not demonstrate a statistically significant improvement in forecasting accuracy compared to naïve baseline approaches such as moving averages.

Alternative Hypothesis (H_{11}): Models using historical traffic data demonstrate a statistically significant improvement in forecasting accuracy compared to naïve baseline approaches.

Research Question 2:

Does the inclusion of real-time traffic data improve congestion forecasting accuracy relative to historical-only models during peak traffic periods?

Null Hypothesis (H_{02}): The inclusion of real-time traffic data does not significantly improve forecasting accuracy compared to historical-only models during peak periods.

Alternative Hypothesis (H_{12}): The inclusion of real-time traffic data significantly improves forecasting accuracy compared to historical-only models during peak periods.

Research Question 3:

What impact do weather conditions have on short-term traffic congestion forecasts in Toronto?

Null Hypothesis (H_{03}): Weather variables such as precipitation and visibility do not have a statistically significant effect on congestion forecasting accuracy.

Alternative Hypothesis (H_{13}): Weather variables such as precipitation and visibility have a statistically significant effect on congestion forecasting accuracy.

Research Question 4:

Can operational optimization and what-if simulations reduce predicted congestion levels at high-traffic intersections?

Null Hypothesis (H_{04}): Operational optimization strategies do not result in a statistically significant reduction in predicted congestion levels.

Alternative Hypothesis (H_{14}): Operational optimization strategies result in a statistically significant reduction in predicted congestion levels.

Proposed Analytical Models:

The proposed analytical approach will explore a combination of time-series forecasting and machine learning techniques. Baseline models will include moving average and ARIMA-based time-series models. Advanced models will include machine learning regression techniques such as Random Forest and Gradient Boosting, as well as deep learning–based spatio-temporal models such as Long Short-Term Memory (LSTM) networks. Model performance will be evaluated using standard forecasting error metrics, including Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

Proposed Data Sources:

City of Toronto Open Data Portal

Historical traffic volume and congestion-related datasets for Toronto will be obtained from the City of Toronto Open Data Portal for model training and historical analysis.

https://open.toronto.ca/

TomTom Traffic Data Services

Near real-time traffic flow and congestion data will be sourced from TomTom traffic services and used as live inputs for short-term traffic prediction.

https://www.tomtom.com/

HERE Technologies Traffic Services

Real-time traffic flow and incident information will be obtained from HERE Technologies to support congestion monitoring and predictive analytics.

https://www.here.com/

OpenWeather Data Services

Real-time and forecasted weather data will be obtained from OpenWeather and incorporated as external factors influencing traffic congestion.

https://openweathermap.org/

All datasets will be cleaned, preprocessed, and temporally aligned to ensure data consistency and analytical reliability.

Table 1: Summary of Proposed Data Sources

Data Source	Data Type	Temporal Nature
Toronto Open Data	Traffic volume	Historical
TomTom / HERE	Traffic flow & congestion	Real-time
OpenWeather	Weather variables	Real-time / Forecast

Intended Outcome:

The intended outcome of this project is the development of a predictive and operational traffic analytics framework that provides actionable insights into congestion patterns in Toronto. The final deliverables will include validated predictive models, operational optimization scenarios, and interactive dashboards that visualize both current and predicted traffic conditions. These outputs can assist transportation planners and policy makers in evaluating intervention strategies, improving traffic flow efficiency, and supporting data-driven urban mobility decisions. The project also demonstrates how advanced data analytics techniques can be applied to real-world smart city transportation challenges and scaled to broader mobility and logistics applications.

References

City of Toronto. (2024). Toronto Open Data portal.

https://open.toronto.ca/

City of Toronto. (2024). Traffic volume and congestion-related datasets.

Toronto Open Data Portal.

https://open.toronto.ca/catalogue/?topics=Transportation

TomTom. (2024). Traffic data and congestion services.

https://www.tomtom.com/

HERE Technologies. (2024). Location and traffic intelligence solutions.

https://www.here.com/

OpenWeather. (2024). Weather data services overview.

https://openweathermap.org/

Zhang, J., Zheng, Y., & Qi, D. (2017).

Deep spatio-temporal residual networks for citywide crowd flows prediction.

Proceedings of the AAAI Conference on Artificial Intelligence, 31(1).

https://ojs.aaai.org/index.php/AAAI/article/view/10735